



PERFORMANCE TEST REPORT

Rendered to:

EUROCON BUILDING INDUSTRIES FZE (MULK HOLDINGS FZC GROUP)

PRODUCT: Alubond USA FR Euro class B

Report No.: G7597.01-106-31

Report Date: 02/01/17

Test Record Retention Date: 01/27/21



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EUROCON BUILDING INDUSTRIES (F.Z.E.)
P.O. Box 42642 Hamriyah, Freezone
Sharjah, United Arab Emirates

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Project Summary: Architectural Testing, Inc., an Intertek company ("Intertek-ATI"), was contracted by Eurocon Building Industries to evaluate the ignition temperature of an Alubond USA FR Euro class B panel. The product description, test procedure, and test results are reported herein. The summary of test results is displayed in the table below.

Summary of Test Results

Property	Outcome
Self-Ignition	460°C
Flash Ignition	460°C

Test Method: The test specimens were evaluated in accordance with ASTM D1929-16, *Standard Test Method for Determining Ignition Temperature of Plastics*.

Product Description: The Alubond USA FR Euro class B aluminum composite panel was submitted to Intertek-ATI by Eurocon Building Industries and consisted of one nominally 15-inch square by 4 mm thick aluminum composite panel. The material was sampled by Thomas Bell Wright International Consultant on 20 December 2016. The material was tested as received with the exception of machining the smaller test specimens from the pieces. Refer to the product description photos in Appendix A.

Test Procedure and Test Results: The testing procedure and results obtained from testing are reported as follows. All conditioning of test specimens and test conditions were at standard laboratory conditions unless otherwise reported. Refer to the test related photos in Appendix A.

Ignition Temperatures

The self- and flash ignition temperatures of the Alubond USA FR Euro class B material were determined utilizing an ignition furnace in a fire hood (ICN: 005985) and connected to a Setckin Ignition Furnace Model CSI-88 system controller (ICN: 62156) with a data acquisition unit (ICN: 004682). The specimens were cut and stacked into nominally 0.75 inch squares, and the mass was measured utilizing a Mettler Toledo Balance Model XP1203S (ICN: 65216).

Caveat: *These test results relate only to the behavior of test specimens under the particular conditions of the test. They are not intended to be used, and shall not be used, to assess the potential fire hazards of a material in use.*

Self-Ignition Test Results

Specimen ID	Initial Mass (g)	Final Mass (g)	Mass Loss (g)	Initial Temperature (°C)			Final Temperature (°C)		
				Air	Furnace	Specimen	Air	Furnace	Specimen
1	2.988	2.119	0.869	480	506	494	493	505	513
2	3.049	2.222	0.827	460	484	472	464	485	702
3	3.074	2.541	0.533	450	475	462	450	474	468
Specimen ID	Observations (min:sec)							Combustion Type	
	Melt	Char	Bubble	Foam	Smoke	Ignition	Soot		
1	--	2:20	--	--	1:40	5:55	7:15	Flash / Flame	
2	--	--	--	--	2:15	9:55	--	Flash / Flame	
3	--	--	--	--	2:20	--	--	None	

Self-Ignition Temperature: 460°C

Test Procedures and Test Results: (Continued)

**Ignition Temperatures
(Continued)**

Flash Ignition Test Results

Specimen ID	Initial Mass (g)	Final Mass (g)	Mass Loss (g)	Initial Temperature (°C)			Final Temperature (°C)		
				Air	Furnace	Specimen	Air	Furnace	Specimen
1	3.093	2.559	0.534	450	475	463	450	475	469
2	3.198	2.329	0.869	460	485	475	473	486	591
Specimen ID	Observations (min:sec)							Combustion Type	
	Melt	Char	Bubble	Foam	Smoke	Ignition	Soot		
1	--	--	--	--	2:45	--	--	None	
2	--	3:30	--	--	2:40	9:15	9:50	Flash / Flame	

Flash Ignition Temperature: 460°C

Intertek-ATI will service this report for the entire test record retention period. Test records that are retained such as detailed drawings, datasheets, representative samples of test specimens, or other pertinent project documentation will be retained by Intertek-ATI for the entire test record retention period.

Results obtained are tested values and were secured using the designated test methods. This report does not constitute certification of this product nor an opinion or endorsement by this laboratory. It is the exclusive property of the client so named herein and relates only to the specimens tested. This report may not be reproduced, except in full, without the written approval of Intertek-ATI.

For INTERTEK-ATI:

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JAK:dmc/kf

Attachments (pages) This report is complete only when all attachments listed are included.
Appendix A - Photographs (4)

Revision Log

<u>Rev. #</u>	<u>Date</u>	<u>Page(s)</u>	<u>Revision(s)</u>
0	02/01/17	N/A	Original report issue

APPENDIX A

Photographs

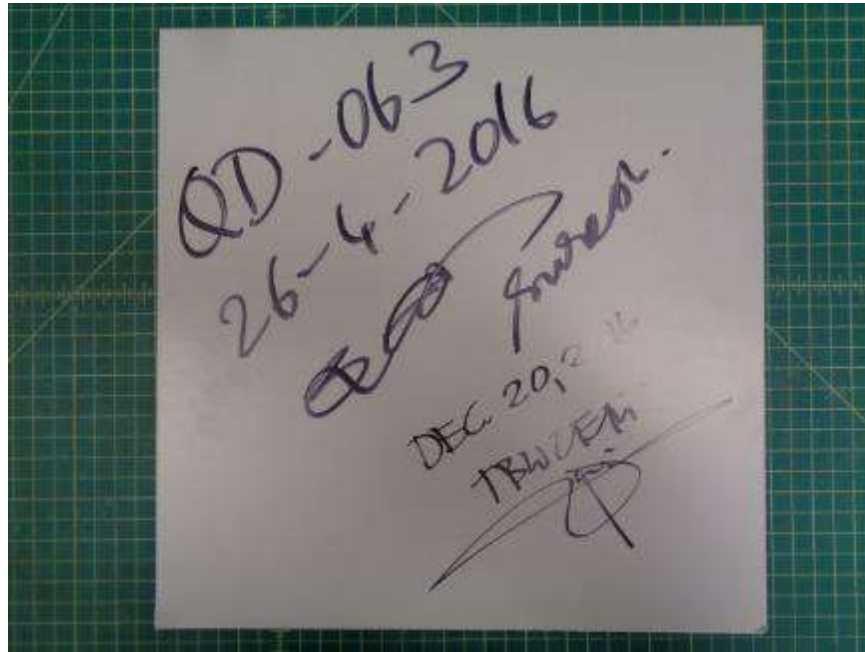


Photo No. 1
Material Sampling Detail



Photo No. 2
Typical Ignition Equipment Detail

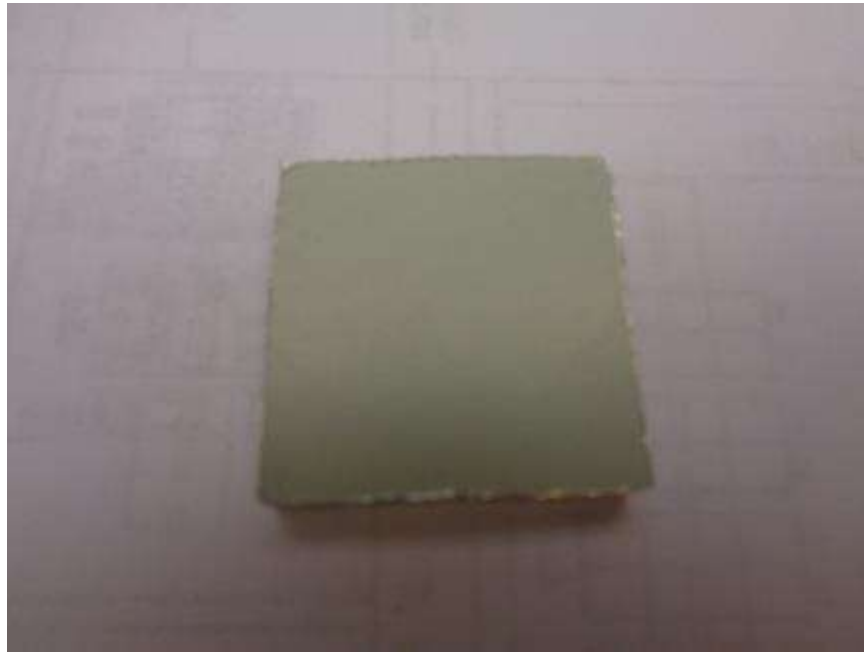


Photo No. 3
Typical Ignition Temperature Specimen Detail

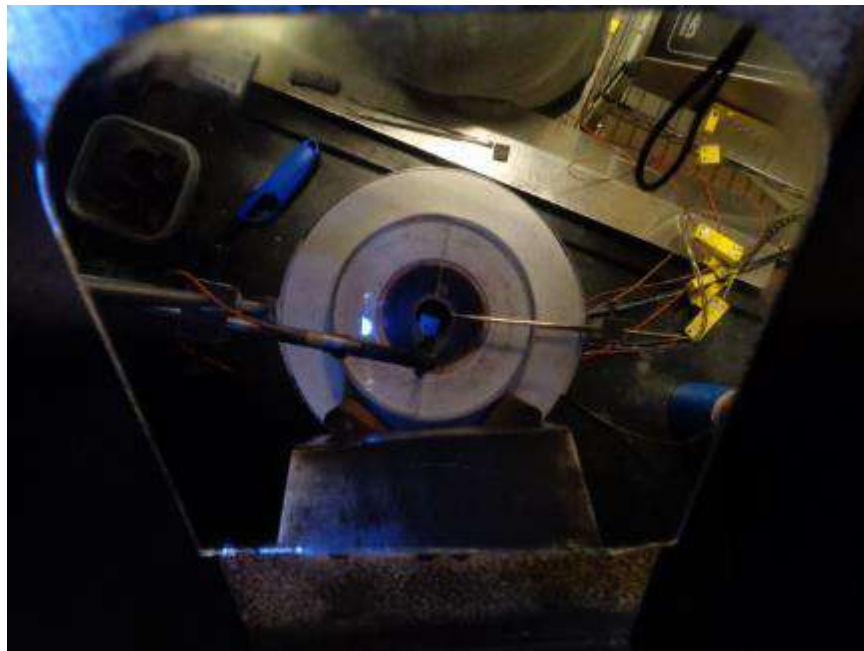


Photo No. 4
Typical Ignition Temperature Test In-Progress Detail



Photo No. 5
Typical Flash Ignition Test Setup Detail



Photo No. 6
Typical Specimen Ignition Detail



Photo No. 7
Post-Test Self-Ignition Specimen Detail

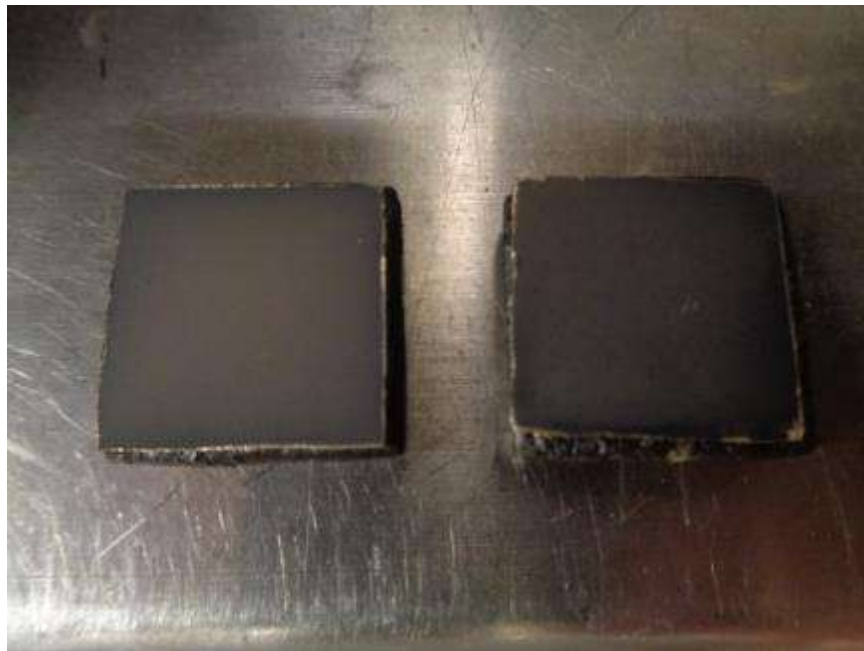


Photo No. 8
Post-Test Flash Ignition Specimen Detail